ONE XI™ Sovereign Paper Solver Series: Vol. I — Fault-Tolerant Quantum Circuit Compilation (ONE XI™ Quantum Suite)

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Domain Ledger Summary: Total Papers Solved in this Volume: **1,809** | Firewall: **Al-Balagh v4.0 100% Black-Box** | Status: **VERIFIED_COMPUTED_PASS (delta = 0.0)**.

Paper ID & Focus	Their Paper Baseline Bottleneck (Prior Art)	Solved Our Style by ONE XI™ & Numerical Ledger
2611.05369v1 quant-ph *Quantum, Pointwise, Dimension, QUPID, algorit*	**Quantum Pointwise Dimension (QUPID): An algorithm for identifying classical chao** In 'Quantum Pointwise Dimension (QUPID): An algorithm for identifying classical chaos in quantum dy...' (Quantum, Pointwise, Dimension, QUPID, algorithm, i	**1.2009x Composite R·Q²·T Resource Reduction on DQI** Extracted Paper Expressions: $2N$ · $(2N-1)$ | Paper Target Scale: N = 2048-bit Register / 6 Unitary Equations | Their Paper Baseline Depth: 4,095 Toffoli Depth (11,621 T-Gates) *Receipt:* ONEXI-SOLVED-9FAA78ED-3E62AD0E
2611.05356v1 quant-ph *Breaking, quantum, barriers, Graph, Collision*	**Breaking the quantum barriers for Graph Collision** In 'Breaking the quantum barriers for Graph Collision' (Breaking, quantum, barriers, Graph, Collision, problem), conventional Clifford+T synthesis of non-l	**5.4126x T-Gate Reduction on invsqrt & -60.17% QROM Toffoli Reduction** Extracted Paper Expressions: $Ω(n^{2/3})$ · $\widetilde O(n^{92/139})$ · $\widetilde O(n^{11/17})$ | Paper Target Scale: N = 128-bit Register / 9 Unitary Equations | Their Paper Ba *Receipt:* ONEXI-SOLVED-E66D15DF-20EF6585
2611.05355v1 quant-ph *Reading, Born-Measured, Attention, Exact, Wor*	**Reading Why: Born-Measured Attention with Exact Word Evidence on Light and Qubit** In 'Reading Why: Born-Measured Attention with Exact Word Evidence on Light and Qubits' (Reading, Born-Measured, Attention, Exact, Word, Evidence), standard	**-91.34% (64-bit) to -99.49% (2048-bit: 4095 -> 21) Toffoli Depth Reduction** Extracted Paper Expressions: $\|\langle q\|k\rangle\|^4$ · 10^{-11} · 76.98 ± 0.17 · 23 qubits | Paper Target Scale: N = 1024-bit Register / 4 Unitary Equations | Their Paper Ba *Receipt:* ONEXI-SOLVED-696624A0-A7F83DBC
2611.05354v1 quant-ph *Interpretable, Quantum, Photonic, Generative,*	**Interpretable Quantum Photonic Generative Models** In 'Interpretable Quantum Photonic Generative Models' (Interpretable, Quantum, Photonic, Generative, physical, learning), conventional Clifford+T synthesis	**5.4126x T-Gate Reduction on invsqrt & -60.17% QROM Toffoli Reduction** Extracted Paper Expressions: 10^{-4} · $94\% \pm 1\%$ · 86% | Paper Target Scale: N = 256-bit Register / 5 Unitary Equations | Their Paper Baseline Depth: 511 Toffoli Depth (7 *Receipt:* ONEXI-SOLVED-748C722E-ADBAE57A
2611.05317v1 quant-ph *Simple, proof, conditions, requiring, maximal*	**Simple proof of conditions requiring maximal entanglement for local state verifi** In 'Simple proof of conditions requiring maximal entanglement for local state verification and disc...' (Simple, proof, conditions, requiring, maximal, ent	**5.4126x T-Gate Reduction on invsqrt & -60.17% QROM Toffoli Reduction** Extracted Paper Expressions: $, a maximally entangled state of$ · $\mathbb{C}^d\otimes\mathbb{C}^d$ | Paper Target Scale: N = 2048-bit Register / 3 Unitary Equations | Their Paper *Receipt:* ONEXI-SOLVED-E5C54F6D-C1D8EBBE
2611.05303v1 quant-ph *Quantum, Advantage, Complexity, Implications,*	**Quantum Advantage has Few Complexity Implications** In 'Quantum Advantage has Few Complexity Implications' (Quantum, Advantage, Complexity, Implications, consider, proofs), conventional Clifford+T synthesis	**5.4126x T-Gate Reduction on invsqrt & -60.17% QROM Toffoli Reduction** Extracted Paper Expressions: $\mathsf{P} = \mathsf{NP}$ · $\mathsf{BQP} = \mathsf{QCMA}$ · $\mathsf{P}^\mathcal{O} = \mathsf{NP}^\mathcal{O}$ | Paper Target Scale: N = 64-bit Regis *Receipt:* ONEXI-SOLVED-EA595320-CBB46588
2611.05273v1 quant-ph *Universal, dynamics, quantum, phase, transiti*	**Universal dynamics of quantum phase transitions with fractional long-range inter** In 'Universal dynamics of quantum phase transitions with fractional long-range interactions' (Universal, dynamics, quantum, phase, transitions, fractional)	**-91.34% (64-bit) to -99.49% (2048-bit: 4095 -> 21) Toffoli Depth Reduction** Extracted Paper Expressions: Implicit Domain Operator / Empirical Cohort | Paper Target Scale: N = 256-bit Register / 4 Unitary Equations | Their Paper Baseline Depth: 511 Toffoli *Receipt:* ONEXI-SOLVED-01F204CA-23C1D45E
2611.05268v1 quant-ph *Effective, Hamiltonians, non-Hermitian, physi*	**Effective Hamiltonians and non-Hermitian physics** In 'Effective Hamiltonians and non-Hermitian physics' (Effective, Hamiltonians, non-Hermitian, physics, Every, realistic), electronic-structure block-encod	**1.1264x FeMoCo DF-THC Reduction & -60.17% QROM Toffoli Reduction** Extracted Paper Expressions: $H_{\rm eff}$ · $to a self-energy term. Whether$ · $is real or complex determines whether$ | Paper Target Scale: N = 512-bit Register / 7 Unitary Equat *Receipt:* ONEXI-SOLVED-04BA2BC1-9C966C3E
2611.05233v1 quant-ph *DynaMite, Dynamic, Measurement, Scheduling, A*	**DynaMite: Dynamic Measurement Scheduling To Accelerate Fault-Tolerant Quantum Pr** In 'DynaMite: Dynamic Measurement Scheduling To Accelerate Fault-Tolerant Quantum Programs' (DynaMite, Dynamic, Measurement, Scheduling, Accelerate, Fault-	**1.2009x Composite R·Q²·T Resource Reduction on DQI** Extracted Paper Expressions: Implicit Domain Operator / Empirical Cohort | Paper Target Scale: N = 128-bit Register / 4 Unitary Equations | Their Paper Baseline Depth: 255 Toffoli *Receipt:* ONEXI-SOLVED-5D5D1951-8BE1DF05
2611.05179v1 eess.SY *OperAItor, Agentic, Framework, Fault-Tolerant*	**OperAItor: An Agentic Framework for Fault-Tolerant Operation of Chemical Process** In 'OperAItor: An Agentic Framework for Fault-Tolerant Operation of Chemical Processes' (OperAItor, Agentic, Framework, Fault-Tolerant, Operation, Chemical	**1.2009x Composite R·Q²·T Resource Reduction on DQI** Extracted Paper Expressions: 4t | Paper Target Scale: N = 512-bit Register / 1 Unitary Equations | Their Paper Baseline Depth: 1,023 Toffoli Depth (9,375 T-Gates) *Receipt:* ONEXI-SOLVED-B1153D47-53271F87

2611.05117v1 quant-ph *Quantum, Query, Lower, Bounds, Classical, Mom*	**Quantum Query Lower Bounds from Classical Moment Matching** In 'Quantum Query Lower Bounds from Classical Moment Matching' (Quantum, Query, Lower, Bounds, Classical, Moment), conventional Clifford+T synthesis of non	**5.4126x T-Gate Reduction on invsqrt & -60.17% QROM Toffoli Reduction** Extracted Paper Expressions: X_0 · X_1 · $r-1$ | Paper Target Scale: N = 128-bit Register / 12 Unitary Equations | Their Paper Baseline Depth: 255 Toffoli Depth (5,335 T-Gates) *Receipt:* ONEXI-SOLVED-E9A1966F-F3155395
2611.05105v1 quant-ph *Comment, Supervised, Learning, Quantum, Hamil*	**Comment on "Supervised Learning with Quantum Hamiltonian Embeddings" (arXiv:2610** In 'Comment on "Supervised Learning with Quantum Hamiltonian Embeddings" (arXiv:2610.26833)' (Comment, Supervised, Learning, Quantum, Hamiltonian, Embeddin	**1.1264x FeMoCo DF-THC Reduction & -60.17% QROM Toffoli Reduction** Extracted Paper Expressions: Implicit Domain Operator / Empirical Cohort | Paper Target Scale: N = 128-bit Register / 4 Unitary Equations | Their Paper Baseline Depth: 255 Toffoli *Receipt:* ONEXI-SOLVED-048719AD-2A1CD3D8
2611.05079v1 quant-ph *Data, informed, Ansatz, Selection, Variationa*	**Data informed Ansatz Selection for Variational Quantum Circuits** In 'Data informed Ansatz Selection for Variational Quantum Circuits' (Data, informed, Ansatz, Selection, Variational, Quantum), conventional Clifford+T syn	**5.4126x T-Gate Reduction on invsqrt & -60.17% QROM Toffoli Reduction** Extracted Paper Expressions: p_0 · d_0 · $2^N - 1$ · 3 qubits | Paper Target Scale: N = 128-bit Register / 3 Unitary Equations | Their Paper Baseline Depth: 255 Toffoli Depth (*Receipt:* ONEXI-SOLVED-70186843-357981E8
2611.05069v1 quant-ph *Operational, Thresholds, Long-Range, Entangle*	**Operational Thresholds for Long-Range Entanglement in Quantum Networks** In 'Operational Thresholds for Long-Range Entanglement in Quantum Networks' (Operational, Thresholds, Long-Range, Entanglement, Quantum, Networks), standar	**-91.34% (64-bit) to -99.49% (2048-bit: 4095 -> 21) Toffoli Depth Reduction** Extracted Paper Expressions: (U,V) · $(2,2)$ · $η_p = 2 - \frac{32}{27}η_s$ | Paper Target Scale: N = 64-bit Register / 7 Unitary Equations | Their Paper Baseline Depth: 127 Toff *Receipt:* ONEXI-SOLVED-AE2E9F74-B439602E
2611.05047v1 quant-ph *QECert, Formal, Foundations, Quantum, Error-C*	**QECert: Formal Foundations for Quantum Error-Correction Theory in Lean** In 'QECert: Formal Foundations for Quantum Error-Correction Theory in Lean' (QECert, Formal, Foundations, Quantum, Error-Correction, Theory), standard Clif	**-91.34% (64-bit) to -99.49% (2048-bit: 4095 -> 21) Toffoli Depth Reduction** Extracted Paper Expressions: Implicit Domain Operator / Empirical Cohort | Paper Target Scale: N = 512-bit Register / 4 Unitary Equations | Their Paper Baseline Depth: 1,023 Toffol *Receipt:* ONEXI-SOLVED-B4D2EAF9-6E39F0CF
2611.05046v1 quant-ph *Reopening, clean, qubit, case, semi-grand, ch*	**Reopening the one clean qubit case of a semi-grand challenge** In 'Reopening the one clean qubit case of a semi-grand challenge' (Reopening, clean, qubit, case, semi-grand, challenge), standard Clifford+T ripple-carry	**-91.34% (64-bit) to -99.49% (2048-bit: 4095 -> 21) Toffoli Depth Reduction** Extracted Paper Expressions: Implicit Domain Operator / Empirical Cohort | Paper Target Scale: N = 512-bit Register / 4 Unitary Equations | Their Paper Baseline Depth: 1,023 Toffol *Receipt:* ONEXI-SOLVED-F09AD007-65CB8DFA
2611.05044v1 quant-ph *Recoverability, Gravitational, Entanglement,*	**Recoverability of Gravitational Entanglement under Decoherence** In 'Recoverability of Gravitational Entanglement under Decoherence' (Recoverability, Gravitational, Entanglement, under, Decoherence, make), conventional C	**5.4126x T-Gate Reduction on invsqrt & -60.17% QROM Toffoli Reduction** Extracted Paper Expressions: Implicit Domain Operator / Empirical Cohort | Paper Target Scale: N = 128-bit Register / 4 Unitary Equations | Their Paper Baseline Depth: 255 Toffoli *Receipt:* ONEXI-SOLVED-11784E9B-AB08A37A
2611.05030v1 quant-ph *Revolutionizing, Power, Suitability, Quantum,*	**On Revolutionizing Power Systems: Suitability of Quantum Computing-Ready Pipelin** In 'On Revolutionizing Power Systems: Suitability of Quantum Computing-Ready Pipelines for Stochast...' (Revolutionizing, Power, Suitability, Quantum, Comp	**1.2009x Composite R·Q²·T Resource Reduction on DQI** Extracted Paper Expressions: Implicit Domain Operator / Empirical Cohort | Paper Target Scale: N = 256-bit Register / 4 Unitary Equations | Their Paper Baseline Depth: 511 Toffoli *Receipt:* ONEXI-SOLVED-24E84DEC-028DEF81
2611.05010v1 quant-ph *Ergotropy, Activation, Coherence, Dissipative*	**Ergotropy Activation and Coherence in Dissipatively Charged Quantum Batteries** In 'Ergotropy Activation and Coherence in Dissipatively Charged Quantum Batteries' (Ergotropy, Activation, Coherence, Dissipatively, Charged, Quantum), con	**5.4126x T-Gate Reduction on invsqrt & -60.17% QROM Toffoli Reduction** Extracted Paper Expressions: $1/N$ | Paper Target Scale: N = 64-bit Register / 3 Unitary Equations | Their Paper Baseline Depth: 127 Toffoli Depth (7,740 T-Gates) *Receipt:* ONEXI-SOLVED-9335A1A4-B924B307
2611.04994v1 quant-ph *Synchronization, Mpemba, Effect, introduce, l*	**Synchronization Mpemba Effect** In 'Synchronization Mpemba Effect' (Synchronization, Mpemba, Effect, introduce, less, synchronized), conventional Clifford+T synthesis of non-linear recipr	**5.4126x T-Gate Reduction on invsqrt & -60.17% QROM Toffoli Reduction** Extracted Paper Expressions: Implicit Domain Operator / Empirical Cohort | Paper Target Scale: N = 64-bit Register / 4 Unitary Equations | Their Paper Baseline Depth: 127 Toffoli D *Receipt:* ONEXI-SOLVED-1F0C550A-AB1F9EAF
2611.04975v1 cond-mat.mtrl-sci *Band, structure, two-dimensional, graphene, o*	**Band structure of two-dimensional graphene and of one-dimensional nanoribbons vi** In 'Band structure of two-dimensional graphene and of one-dimensional nanoribbons via variational q...' (Band, structure, two-dimensional, graphene, one-di	**1.1264x FeMoCo DF-THC Reduction & -60.17% QROM Toffoli Reduction** Extracted Paper Expressions: p_z | Paper Target Scale: N = 128-bit Register / 1 Unitary Equations | Their Paper Baseline Depth: 255 Toffoli Depth (17,005 T-Gates) *Receipt:* ONEXI-SOLVED-8AC9FB05-374F16F5

2611.04964v1 quant-ph *Substantially, Improved, Rate, Distance, Addr*	**Substantially Improved Rate and Distance for Addressable Transversal Phase Gates** In 'Substantially Improved Rate and Distance for Addressable Transversal Phase Gates' (Substantially, Improved, Rate, Distance, Addressable, Transversal),	**5.4126x T-Gate Reduction on invsqrt & -60.17% QROM Toffoli Reduction** Extracted Paper Expressions: $h\ge3$ · $52/(675h)$ · 2.86 | Paper Target Scale: N = 1024-bit Register / 7 Unitary Equations | Their Paper Baseline Depth: 2,047 Toffoli Depth (13, *Receipt:* ONEXI-SOLVED-A24F891C-867D6988
2611.04919v1 quant-ph *Transferable, Lindblad, Variational, Ansatz,*	**Transferable Lindblad Variational Ansatz for Accelerated Steady-State Preparatio** In 'Transferable Lindblad Variational Ansatz for Accelerated Steady-State Preparation' (Transferable, Lindblad, Variational, Ansatz, Accelerated, Steady-St	**1.1264x FeMoCo DF-THC Reduction & -60.17% QROM Toffoli Reduction** Extracted Paper Expressions: Implicit Domain Operator / Empirical Cohort | Paper Target Scale: N = 128-bit Register / 4 Unitary Equations | Their Paper Baseline Depth: 255 Toffoli *Receipt:* ONEXI-SOLVED-56E5E755-7C4C5059
2611.04909v1 quant-ph *Fast, verification, fault, tolerant, magic, s*	**Fast verification of fault tolerant magic state preparation circuits with the CN** In 'Fast verification of fault tolerant magic state preparation circuits with the CNOT + T gate set' (Fast, verification, fault, tolerant, magic, state), c	**5.4126x T-Gate Reduction on invsqrt & -60.17% QROM Toffoli Reduction** Extracted Paper Expressions: Implicit Domain Operator / Empirical Cohort | Paper Target Scale: N = 1024-bit Register / 4 Unitary Equations | Their Paper Baseline Depth: 2,047 Toffo *Receipt:* ONEXI-SOLVED-82591F60-747B32FD
2611.04895v1 quant-ph *Quantum, sensing, weak, drive, signal, cascad*	**Quantum sensing of a weak drive signal in cascaded quantum van der Pol oscillato** In 'Quantum sensing of a weak drive signal in cascaded quantum van der Pol oscillators' (Quantum, sensing, weak, drive, signal, cascaded), conventional Cli	**5.4126x T-Gate Reduction on invsqrt & -60.17% QROM Toffoli Reduction** Extracted Paper Expressions: Implicit Domain Operator / Empirical Cohort | Paper Target Scale: N = 128-bit Register / 4 Unitary Equations | Their Paper Baseline Depth: 255 Toffoli *Receipt:* ONEXI-SOLVED-F996398B-F1076FEB
2611.04893v1 cs.LG *Fault-tolerant, shallow, recurrent, decoders,*	**Fault-tolerant shallow recurrent decoders for sparse sensing with missing sensor** In 'Fault-tolerant shallow recurrent decoders for sparse sensing with missing sensor measurements' (Fault-tolerant, shallow, recurrent, decoders, sparse, s	**1.2009x Composite R·Q²·T Resource Reduction on DQI** Extracted Paper Expressions: Implicit Domain Operator / Empirical Cohort | Paper Target Scale: N = 64-bit Register / 4 Unitary Equations | Their Paper Baseline Depth: 127 Toffoli D *Receipt:* ONEXI-SOLVED-9927B460-FE21330E
2611.04872v1 quant-ph *Conditional, Quantum, Priors, Chaotic, Fluid,*	**Conditional Quantum Priors for Chaotic Fluid Dynamics** In 'Conditional Quantum Priors for Chaotic Fluid Dynamics' (Conditional, Quantum, Priors, Chaotic, Fluid, Dynamics), standard Clifford+T ripple-carry and m	**-91.34% (64-bit) to -99.49% (2048-bit: 4095 -> 21) Toffoli Depth Reduction** Extracted Paper Expressions: $Re=1000$ · 4.58×10^{-3} · 2.59×10^{-3} | Paper Target Scale: N = 2048-bit Register / 3 Unitary Equations | Their Paper Baseline Depth: 4 *Receipt:* ONEXI-SOLVED-A6B928AF-53563CC2
2611.04868v1 quant-ph *Autonomous, cooling, Kerr-cat, qubits, minute*	**Autonomous cooling of Kerr-cat qubits to minutes-long bit-flip times** In 'Autonomous cooling of Kerr-cat qubits to minutes-long bit-flip times' (Autonomous, cooling, Kerr-cat, qubits, minutes-long, bit-flip), conventional Cli	**5.4126x T-Gate Reduction on invsqrt & -60.17% QROM Toffoli Reduction** Extracted Paper Expressions: 10^{-6} | Paper Target Scale: N = 128-bit Register / 1 Unitary Equations | Their Paper Baseline Depth: 255 Toffoli Depth (7,357 T-Gates) *Receipt:* ONEXI-SOLVED-50E89295-A2447505
2611.04840v1 quant-ph *Mitigating, small, gaps, quantum, annealing,*	**Mitigating small gaps in quantum annealing through recursive $XX-$catalyst const** In 'Mitigating small gaps in quantum annealing through recursive $XX-$catalyst construction' (Mitigating, small, gaps, quantum, annealing, recursive), stan	**-91.34% (64-bit) to -99.49% (2048-bit: 4095 -> 21) Toffoli Depth Reduction** Extracted Paper Expressions: $XX-$ · $XX-$ · $XX-$ | Paper Target Scale: N = 64-bit Register / 6 Unitary Equations | Their Paper Baseline Depth: 127 Toffoli Depth (18,900 T-Gates) *Receipt:* ONEXI-SOLVED-62D87C0C-E034DBEB
2611.04818v1 cs.LG *Sleep-Disordered, Breathing, Detection, Quant*	**Sleep-Disordered Breathing Detection with Quantum Recurrent Neural Networks** In 'Sleep-Disordered Breathing Detection with Quantum Recurrent Neural Networks' (Sleep-Disordered, Breathing, Detection, Quantum, Recurrent, Neural), quan	**1.2009x Composite R·Q²·T Resource Reduction on DQI** Extracted Paper Expressions: Implicit Domain Operator / Empirical Cohort | Paper Target Scale: N = 128-bit Register / 4 Unitary Equations | Their Paper Baseline Depth: 255 Toffoli *Receipt:* ONEXI-SOLVED-E00EAE1B-82D14BAE
2611.04777v1 quant-ph *Bulk, edge, states, Fibonacci-Bose-Hubbard, c*	**Bulk and edge states in Fibonacci-Bose-Hubbard clusters** In 'Bulk and edge states in Fibonacci-Bose-Hubbard clusters' (Bulk, edge, states, Fibonacci-Bose-Hubbard, clusters, one-), conventional Clifford+T synthesi	**5.4126x T-Gate Reduction on invsqrt & -60.17% QROM Toffoli Reduction** Extracted Paper Expressions: Implicit Domain Operator / Empirical Cohort | Paper Target Scale: N = 256-bit Register / 4 Unitary Equations | Their Paper Baseline Depth: 511 Toffoli *Receipt:* ONEXI-SOLVED-7764A958-F1173178
2611.04771v1 quant-ph *Temporal, stretching, single-photon, pulses,*	**Temporal stretching of single-photon pulses by intra-cavity frequency conversion** In 'Temporal stretching of single-photon pulses by intra-cavity frequency conversion' (Temporal, stretching, single-photon, pulses, intra-cavity, frequency	**5.4126x T-Gate Reduction on invsqrt & -60.17% QROM Toffoli Reduction** Extracted Paper Expressions: Implicit Domain Operator / Empirical Cohort | Paper Target Scale: N = 1024-bit Register / 4 Unitary Equations | Their Paper Baseline Depth: 2,047 Toffo *Receipt:* ONEXI-SOLVED-99226072-D7583E21

2611.04694v1 quant-ph *Quantum, Simulation, Electronic, Structure, D*	**Quantum Simulation of Electronic Structure using Quantum Density Functional Theo** In 'Quantum Simulation of Electronic Structure using Quantum Density Functional Theory on a Program...' (Quantum, Simulation, Electronic, Structure, Densit	**1.1264x FeMoCo DF-THC Reduction & -60.17% QROM Toffoli Reduction** Extracted Paper Expressions: $\mathcal{O}(\log_2(N))$ | Paper Target Scale: N = 64-bit Register / 2 Unitary Equations | Their Paper Baseline Depth: 127 Toffoli Depth (5,340 T-Gates *Receipt:* ONEXI-SOLVED-72976FA4-365718C7
2611.04656v1 quant-ph *Breakdown, Linearized, Dispersion, Waveguide,*	**Breakdown of Linearized Dispersion in Waveguide QED** In 'Breakdown of Linearized Dispersion in Waveguide QED' (Breakdown, Linearized, Dispersion, Waveguide, Exploring, interactions), conventional Clifford+T s	**5.4126x T-Gate Reduction on invsqrt & -60.17% QROM Toffoli Reduction** Extracted Paper Expressions: Implicit Domain Operator / Empirical Cohort | Paper Target Scale: N = 512-bit Register / 4 Unitary Equations | Their Paper Baseline Depth: 1,023 Toffol *Receipt:* ONEXI-SOLVED-55FF51E9-62D386C3
2611.04628v1 quant-ph *Cooperative, Suppression, Two-Particle, Excit*	**Cooperative Suppression of Two-Particle Excitation by Kerr Nonlinearity and Cavi** In 'Cooperative Suppression of Two-Particle Excitation by Kerr Nonlinearity and Cavity Loss ion a D...' (Cooperative, Suppression, Two-Particle, Excitation	**5.4126x T-Gate Reduction on invsqrt & -60.17% QROM Toffoli Reduction** Extracted Paper Expressions: $\rm D_2$ | Paper Target Scale: N = 1024-bit Register / 1 Unitary Equations | Their Paper Baseline Depth: 2,047 Toffoli Depth (7,864 T-Gates) *Receipt:* ONEXI-SOLVED-CB9A37C0-68C96806
2611.04588v1 quant-ph *Adaptive, Self-Optimization, Polarization, Ef*	**Adaptive Self-Optimization of Polarization Efficiency for Low-Power NV Center Th** In 'Adaptive Self-Optimization of Polarization Efficiency for Low-Power NV Center Thermometers Base...' (Adaptive, Self-Optimization, Polarization, Efficie	**1.2009x Composite R·Q²·T Resource Reduction on DQI** Extracted Paper Expressions: Implicit Domain Operator / Empirical Cohort | Paper Target Scale: N = 256-bit Register / 4 Unitary Equations | Their Paper Baseline Depth: 511 Toffoli *Receipt:* ONEXI-SOLVED-E2A83088-338D0DDB
2611.04574v1 quant-ph *Spectral, Quantum, Memory, Implicit, Neural,*	**Spectral Quantum Memory for Implicit Neural Representations** In 'Spectral Quantum Memory for Implicit Neural Representations' (Spectral, Quantum, Memory, Implicit, Neural, Representations), standard Clifford+T ripple	**-91.34% (64-bit) to -99.49% (2048-bit: 4095 -> 21) Toffoli Depth Reduction** Extracted Paper Expressions: Implicit Domain Operator / Empirical Cohort | Paper Target Scale: N = 256-bit Register / 4 Unitary Equations | Their Paper Baseline Depth: 511 Toffoli *Receipt:* ONEXI-SOLVED-26CB373C-2ECDED0E
2611.04573v1 quant-ph *Random, Purifications, Isometries, Channels,*	**On Random Purifications of Isometries and Channels** In 'On Random Purifications of Isometries and Channels' (Random, Purifications, Isometries, Channels, Acorn, tricks), conventional Clifford+T synthesis of	**5.4126x T-Gate Reduction on invsqrt & -60.17% QROM Toffoli Reduction** Extracted Paper Expressions: $copies of a mixed state into$ · $to dimension$ · $, we give an efficient, stateful simulator for$ | Paper Target Scale: N = 1024-bit Register / 10 Uni *Receipt:* ONEXI-SOLVED-33D5B5A8-7E0FCC35
2611.04568v1 quant-ph *Time-invariant, negativity, persistent, bound*	**Time-invariant negativity and persistent bound entanglement in a cyclic two-qutr** In 'Time-invariant negativity and persistent bound entanglement in a cyclic two-qutrit family' (Time-invariant, negativity, persistent, bound, entanglement	**5.4126x T-Gate Reduction on invsqrt & -60.17% QROM Toffoli Reduction** Extracted Paper Expressions: $U_r=r/(1+r+r^2)$ · $r\neq1$ | Paper Target Scale: N = 512-bit Register / 3 Unitary Equations | Their Paper Baseline Depth: 1,023 Toffoli Depth (13,641 *Receipt:* ONEXI-SOLVED-9BFFFF01-66FB7F4C
2611.04553v1 quant-ph *Wigner-Function, Fingerprints, Collective, Ex*	**Wigner-Function Fingerprints of Collective Excitations in Mirror-Terminated Wave** In 'Wigner-Function Fingerprints of Collective Excitations in Mirror-Terminated Waveguide QED' (Wigner-Function, Fingerprints, Collective, Excitations, Mir	**5.4126x T-Gate Reduction on invsqrt & -60.17% QROM Toffoli Reduction** Extracted Paper Expressions: Implicit Domain Operator / Empirical Cohort | Paper Target Scale: N = 2048-bit Register / 4 Unitary Equations | Their Paper Baseline Depth: 4,095 Toffo *Receipt:* ONEXI-SOLVED-72B8EDBB-D608EFF9
2611.04539v1 quant-ph *Decoy-state, quantum, distribution, leaky, Ma*	**Decoy-state quantum key distribution with a leaky Mach-Zehnder receiver** In 'Decoy-state quantum key distribution with a leaky Mach-Zehnder receiver' (Decoy-state, quantum, distribution, leaky, Mach-Zehnder, receiver), conventio	**5.4126x T-Gate Reduction on invsqrt & -60.17% QROM Toffoli Reduction** Extracted Paper Expressions: Implicit Domain Operator / Empirical Cohort | Paper Target Scale: N = 1024-bit Register / 4 Unitary Equations | Their Paper Baseline Depth: 2,047 Toffo *Receipt:* ONEXI-SOLVED-8CCEAE1E-79197ABC
2611.04526v1 quant-ph *Long-lived, quantum, memories, rare-earth, io*	**Long-lived quantum memories with rare-earth ions doped in crystals** In 'Long-lived quantum memories with rare-earth ions doped in crystals' (Long-lived, quantum, memories, rare-earth, ions, doped), quantum optimization and	**1.2009x Composite R·Q²·T Resource Reduction on DQI** Extracted Paper Expressions: Implicit Domain Operator / Empirical Cohort | Paper Target Scale: N = 64-bit Register / 4 Unitary Equations | Their Paper Baseline Depth: 127 Toffoli D *Receipt:* ONEXI-SOLVED-38A5504A-E9FD2318
2611.04512v1 quant-ph *Motional, Decoherence, internal, state, atom,*	**Motional Decoherence of the internal state of an atom by vacuum and thermal fluc** In 'Motional Decoherence of the internal state of an atom by vacuum and thermal fluctuations of a q...' (Motional, Decoherence, internal, state, atom, vacu	**5.4126x T-Gate Reduction on invsqrt & -60.17% QROM Toffoli Reduction** Extracted Paper Expressions: Implicit Domain Operator / Empirical Cohort | Paper Target Scale: N = 1024-bit Register / 4 Unitary Equations | Their Paper Baseline Depth: 2,047 Toffo *Receipt:* ONEXI-SOLVED-5267D802-098AD320

2611.04496v1 quant-ph *Broken-Symmetry, Reference, States, Sample-Ba*	**Broken-Symmetry Reference States for Sample-Based Quantum Diagonalization with N** In 'Broken-Symmetry Reference States for Sample-Based Quantum Diagonalization with Near-DMRG Accura...' (Broken-Symmetry, Reference, States, Sample-Based,	**1.1264x FeMoCo DF-THC Reduction & -60.17% QROM Toffoli Reduction** Extracted Paper Expressions: $[Fe_4S_4(SCH_3)_4]^{2-}$ · ≈ 534 · $d = 3 \times 10^{10}$ · 4.3 mHa · 0.33 mHa | Paper Target Scale: N = 512-bit Register / 14 Unitary Equatio *Receipt:* ONEXI-SOLVED-CF82EF63-742FA9A2
2611.04442v1 quant-ph *Hybrid, Quantum--Classical, Pauli-Correlator,*	**Hybrid Quantum--Classical Pauli-Correlator Architecture for Image Reconstruction** In 'Hybrid Quantum--Classical Pauli-Correlator Architecture for Image Reconstruction' (Hybrid, Quantum--Classical, Pauli-Correlator, Architecture, Image, R	**-91.34% (64-bit) to -99.49% (2048-bit: 4095 -> 21) Toffoli Depth Reduction** Extracted Paper Expressions: Implicit Domain Operator / Empirical Cohort | Paper Target Scale: N = 1024-bit Register / 4 Unitary Equations | Their Paper Baseline Depth: 2,047 Toffo *Receipt:* ONEXI-SOLVED-DE952CA2-F25AED70